

A2-Level Maths:

Core 4

for Edexcel



C4.6 Integration 2



These icons indicate that teacher's notes or useful web addresses are available in the Notes Page.

This icon indicates the slide contains activities created in Flash. These activities are not editable.

For more detailed instructions, see the *Getting Started* presentation.

- **Using trigonometric identities in integration**
- Using partial fractions in integration
- First-order differential equations
- Differential equations with separable variables
- Using differential equations to model real-life situations
- The trapezium rule
- Examination-style questions



Using trigonometric identities in integration

Many expressions involving trigonometric functions cannot be integrated directly using standard integrals.

In these cases, it may be possible to rewrite the expression using an appropriate trigonometric identity.

For example:

Find $\int \sin x \cos x \, dx$.

Using the double angle formula for $\sin 2x$:

$$\sin 2x \equiv 2 \sin x \cos x$$

So, we can write:

$$\begin{aligned} \int \sin x \cos x \, dx &= \frac{1}{2} \int \sin 2x \, dx \\ &= -\frac{1}{4} \cos 2x + c \end{aligned}$$

Integrating $\cos^2 x$ and $\sin^2 x$

To integrate functions involving even powers of $\cos x$ and $\sin x$ we can use the double angle formulae for $\cos 2x$.

There are two ways of writing this involving $\sin^2 x$ and $\cos^2 x$:

$$\cos 2x \equiv 2\cos^2 x - 1$$

$$\cos 2x \equiv 1 - 2\sin^2 x$$

We can rewrite these with $\sin^2 x$ and $\cos^2 x$ as the subject:

$$\cos^2 x \equiv \frac{1}{2}(1 + \cos 2x)$$

①

$$\sin^2 x \equiv \frac{1}{2}(1 - \cos 2x)$$

②



Integrating $\cos^2 x$ and $\sin^2 x$

Find $\int \cos^2 x \, dx$.

Using ①

$$\begin{aligned}\int \cos^2 x \, dx &= \frac{1}{2} \int (1 + \cos 2x) \, dx \\ &= \frac{1}{2} \left(x + \frac{1}{2} \sin 2x \right) + c\end{aligned}$$

Find $\int \sin^2 2x \, dx$.

Using ② and replacing x with $2x$ gives:

$$\begin{aligned}\int \sin^2 2x \, dx &= \frac{1}{2} \int (1 - \cos 4x) \, dx \\ &= \frac{1}{2} \left(x - \frac{1}{4} \sin 4x \right) + c\end{aligned}$$



Integrating even powers of $\cos x$ and $\sin x$

We can extend the use of these identities to integrate any even power of $\cos x$ or $\sin x$. For example:

Find $\int \cos^4 \frac{1}{2} x \, dx$.

This can be written in terms of $\cos^2 \frac{1}{2} x$ as:

$$\begin{aligned}\int \cos^4 \frac{1}{2} x \, dx &= \int (\cos^2 \frac{1}{2} x)^2 \, dx \\ &= \int \left(\frac{1}{2}(1 + \cos x)\right)^2 \, dx \\ &= \frac{1}{4} \int (1 + 2\cos x + \cos^2 x) \, dx \\ &= \frac{1}{4} \int \left(1 + 2\cos x + \frac{1}{2}(1 + \cos 2x)\right) \, dx \\ &= \frac{1}{4} \int \left(\frac{3}{2} + 2\cos x + \frac{1}{2}\cos 2x\right) \, dx \\ &= \frac{1}{4} \left(\frac{3}{2}x + 2\sin x + \frac{1}{4}\sin 2x\right) + c\end{aligned}$$

Integrating odd powers of $\cos x$ and $\sin x$

Odd powers of $\cos x$ and $\sin x$ can be integrated using the identity $\cos^2 x + \sin^2 x = 1$.

$$\cos^2 x \equiv \frac{1}{2}(1 + \cos 2x) \quad \textcircled{1}$$

$$\sin^2 x \equiv \frac{1}{2}(1 - \cos 2x) \quad \textcircled{2}$$

Find $\int \sin^3 x \, dx$.

$$\int \sin^3 x \, dx = \int \sin^2 x \sin x \, dx$$

Using $\textcircled{2}$

$$= \int (1 - \cos^2 x) \sin x \, dx$$

$$= \int (\sin x - \cos^2 x \sin x) \, dx$$



Integrating odd powers of $\cos x$ and $\sin x$

This is now in a form that we can integrate.

The first part, $\sin x$, integrates to give $-\cos x$.

The second part, $\cos^2 x \sin x$, can be recognized as the product of two functions.

Remember the chain rule for differentiation:

$$y = \square^n \Rightarrow \frac{dy}{dx} = n \square^{n-1} \blacksquare$$

where $\square$ is $f(x)$ and $\blacksquare$ is $f'(x)$.

The derivative of $\cos x$ is $-\sin x$ and so:

$$\frac{d}{dx}(\cos^3 x) = -3\cos^2 x \sin x$$

Integrating odd powers of $\cos x$ and $\sin x$

$$\text{Therefore, } \int \cos^2 x \sin x \, dx = -\frac{1}{3} \cos^3 x + c$$

So, returning to the original problem:

$$\begin{aligned} \int \sin^3 x \, dx &= \int (\sin x - \cos^2 x \sin x) \, dx \\ &= -\cos x + \frac{1}{3} \cos^3 x + c \\ &= \frac{1}{3} \cos x (\cos^2 x - 3) + c \end{aligned}$$



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We have seen that rational functions of the form $\frac{f'(x)}{f(x)}$ can be integrated using:

$$\int \frac{f'(x)}{f(x)} dx = \ln|f(x)|$$

In particular, if $f(x)$ is a linear function then:

$$\int \frac{1}{ax+b} dx = \frac{1}{a} \ln|ax+b| + c$$

Suppose we want to integrate a rational function with more than one linear factor in the denominator.

For example: $\int \frac{2x-1}{(x-2)(x-1)} dx$



Using partial fractions in integration

We can integrate this by first splitting $\frac{2x-1}{(x-2)(x-1)}$ into partial fractions.

Let
$$\frac{2x-1}{(x-2)(x-1)} \equiv \frac{A}{x-2} + \frac{B}{x-1}$$

Multiplying through by $(x-2)(x-1)$:

$$2x-1 \equiv A(x-1) + B(x-2) \quad \textcircled{1}$$

Substituting $x = 2$ into $\textcircled{1}$:

$$4-1 = A$$

$$A = 3$$

Substituting $x = 1$ into $\textcircled{1}$:

$$2-1 = -B$$

$$B = -1$$

Using partial fractions in integration

$$\text{So, } \frac{2x-1}{(x-2)(x-1)} \equiv \frac{3}{x-2} - \frac{1}{x-1}$$

We can now integrate:

$$\begin{aligned} \int \frac{2x-1}{(x-2)(x-1)} dx &= \int \frac{3}{x-2} dx - \int \frac{1}{x-1} dx \\ &= 3\ln|x-2| - \ln|x-1| + c \\ &= \ln|(x-2)^3| - \ln|x-1| + c \\ &= \ln \left| \frac{(x-2)^3}{x-1} \right| + c \end{aligned}$$

Using partial fractions in integration

Find $\int \frac{6}{4x^2 - 9} dx$.

The denominator involves the difference between two squares and so we can write:

$$\int \frac{6}{4x^2 - 9} dx \equiv \int \frac{6}{(2x + 3)(2x - 3)} dx$$

Let
$$\frac{6}{(2x + 3)(2x - 3)} \equiv \frac{A}{2x + 3} + \frac{B}{2x - 3}$$

Multiplying through by $(2x + 3)(2x - 3)$:

$$6 \equiv A(2x - 3) + B(2x + 3) \quad \textcircled{1}$$

Using partial fractions in integration

Substituting $x = -\frac{3}{2}$ into ①:

$$6 = A(2(-\frac{3}{2}) - 3)$$

$$6 = -6A$$

$$A = -1$$

Substituting $x = \frac{3}{2}$ into ①:

$$6 = B(2(\frac{3}{2}) + 3)$$

$$6 = 6B$$

$$B = 1$$

$$\text{So, } \frac{6}{4x^2 - 9} \equiv \frac{1}{2x - 3} - \frac{1}{2x + 3}$$

Using partial fractions in integration

We can now integrate:

$$\begin{aligned}\int \frac{6}{4x^2 - 9} dx &= \int \frac{1}{2x - 3} dx - \int \frac{1}{2x + 3} dx \\ &= \frac{1}{2} \ln|2x - 3| - \frac{1}{2} \ln|2x + 3| + c \\ &= \frac{1}{2} \ln \left| \frac{2x - 3}{2x + 3} \right| + c\end{aligned}$$



Using partial fractions in integration

Find $\int \frac{8x^2 + 21x + 13}{(2x + 1)(x + 2)^2} dx.$

Let $\frac{8x^2 + 21x + 13}{(2x + 1)(x + 2)^2} \equiv \frac{A}{2x + 1} + \frac{B}{x + 2} + \frac{C}{(x + 2)^2}$

Multiplying through by $(2x + 1)(x + 2)^2$:

$$8x^2 + 21x + 13 \equiv A(x + 2)^2 + B(2x + 1)(x + 2) + C(2x + 1) \quad \textcircled{1}$$

Substituting $x = -\frac{1}{2}$ into $\textcircled{1}$:

$$2 - \frac{21}{2} + 13 = \frac{9A}{4}$$

$$\frac{9}{2} = \frac{9A}{4}$$

$$A = 2$$

Using partial fractions in integration

Substituting $x = -2$ into ①:

$$32 - 42 + 13 = -3C$$

$$3 = -3C$$

$$C = -1$$

Comparing the coefficients of x^2 :

$$8 = A + 2B$$

$$8 = 2 + 2B$$

$$B = 3$$

$$\text{So, } \frac{8x^2 + 21x + 13}{(2x + 1)(x + 2)^2} \equiv \frac{2}{2x + 1} + \frac{3}{x + 2} - \frac{1}{(x + 2)^2}$$

Using partial fractions in integration

We can now integrate:

$$\int \frac{8x^2 + 21x + 13}{(2x+1)(x+2)^2} dx = \int \frac{2}{2x+1} dx + \int \frac{3}{x+2} dx - \int \frac{1}{(x+2)^2} dx$$

$$= \ln|2x+1| + 3\ln|x+2| + \frac{1}{x+2} + c$$

← The integral of $(x+2)^{-2}$ is $-(x+2)^{-1}$.

$$= \ln|2x+1| + \ln|x+2|^3 + \frac{1}{x+2} + c$$

$$= \ln|(2x+1)(x+2)^3| + \frac{1}{x+2} + c$$



Using partial fractions in integration

To integrate an improper fraction we need to rewrite it in proper form before integrating. For example:

Find $\int \frac{4x}{2x+3} dx$.

The improper fraction in this example can be written in proper form by rewriting the numerator.

$$\begin{aligned}\frac{4x}{2x+3} &= \frac{2(2x+3) - 6}{2x+3} \\ &= \frac{2(2x+3)}{2x+3} - \frac{6}{2x+3} \\ &= 2 - \frac{6}{2x+3}\end{aligned}$$

Using partial fractions in integration

We can now integrate:

$$\begin{aligned}\int \frac{4x}{2x+3} dx &= \int 2 dx - \int \frac{6}{2x+3} dx \\ &= 2x - \frac{6}{2} \ln|2x+3| + c \\ &= 2x - 3\ln|2x+3| + c\end{aligned}$$

More difficult examples may require us to set up an inequality or use polynomial long division. For example:

Find $\int \frac{4x^3 + 10x + 4}{2x^2 + x} dx$.

Let $\frac{4x^3 + 10x + 4}{x(2x+1)} \equiv Ax + B + \frac{C}{x} + \frac{D}{2x+1}$

Using partial fractions in integration

Multiplying through by $x(2x + 1)$:

$$4x^3 + 10x + 4 \equiv (Ax + B)x(2x + 1) + C(2x + 1) + Dx \quad \textcircled{1}$$

Substituting $x = 0$ into $\textcircled{1}$:

$$4 = C$$

$$C = 4$$

Substituting $x = -\frac{1}{2}$ into $\textcircled{1}$:

$$-\frac{4}{8} - 5 + 4 = -\frac{D}{2}$$

$$-\frac{3}{2} = -\frac{D}{2}$$

$$D = 3$$

A and B can now be found by comparing coefficients.



Using partial fractions in integration

Comparing coefficients of x^3 :

$$4 = 2A$$

$$A = 2$$

Comparing coefficients of x^2 :

$$0 = A + 2B$$

$$-2B = 2$$

$$B = -1$$

So,
$$\frac{4x^3 + 10x + 4}{x(2x + 1)} \equiv 2x - 1 + \frac{4}{x} + \frac{3}{2x + 1}$$



Using partial fractions in integration

We can now integrate:

$$\int \frac{4x^3 + 10x + 4}{x(2x+1)} dx = \int (2x-1)dx + \int \frac{4}{x} dx + \int \frac{3}{2x+1} dx$$

$$= x^2 - x + 4\ln|x| + \frac{3}{2}\ln|2x+1| + c$$

$$= x^2 - x + \ln|x^4| + \ln|(2x+1)^{\frac{3}{2}}| + c$$

$$= x^2 - x + \ln|x^4(2x+1)^{\frac{3}{2}}| + c$$

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A **differential equation** in two variables x and y is an equation that contains derivatives of y with respect to x . For example:

$$\frac{dy}{dx} = 4x^2 + 1,$$

$$\frac{dy}{dx} = xy^3,$$

$$2x + y \frac{dy}{dx} = 5xy.$$

The **order** of a differential equation is given by the highest order of derivative that occurs in it.

First-order differential equations contain terms in $\frac{dy}{dx}$,

second-order differential equations contain terms in $\frac{d^2y}{dx^2}$,

third-order differential equations contain terms in $\frac{d^3y}{dx^3}$, etc.

In this course we will only be looking at first-order differential equations.



The solution to a differential equation in x and y will take the form $y = f(x)$.

The simplest differential equations are those of the form:

$$\frac{dy}{dx} = f(x)$$

Differential equations of this form can be solved by integrating both sides with respect to x to give:

$$y = \int f(x) dx$$

For example, suppose we have the differential equation:

$$\frac{dy}{dx} = 4x + 1$$





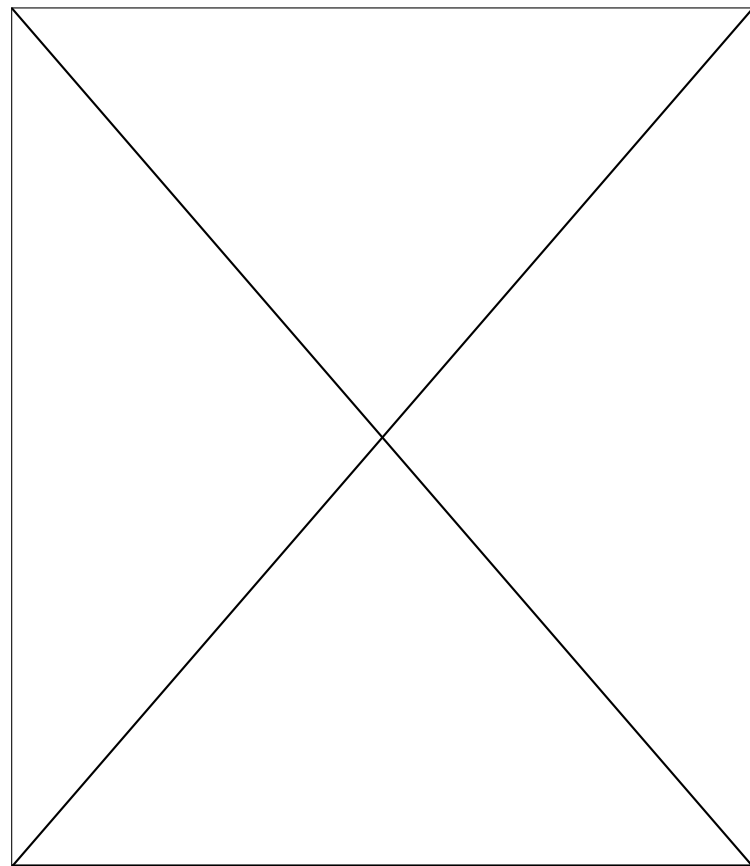
Integrating both sides with respect to x gives:

$$y = \int (4x + 1) dx$$

$$y = 2x^2 + x + c$$

This is called the **general solution** to the equation $\frac{dy}{dx} = 4x + 1$.

Since the constant c can take any value, this represents a whole family of solutions as shown here:



Finding a particular solution

Suppose that as well as being given the differential equation:

$$\frac{dy}{dx} = 4x + 1$$

we are also told that when $x = 1$, $y = 4$.

We can use this additional information to find the value of the arbitrary constant c in the general solution:

$$y = 2x^2 + x + c$$

Substituting $x = 1$ and $y = 4$:

$$4 = 2 + 1 + c$$

$$c = 1$$

This gives us the **particular solution**:

$$y = 2x^2 + x + 1$$



Solving first-order differential equations

Find the particular solution to the differential equation

$$(x^2 + 1) \frac{dy}{dx} = 4x$$

given that $y = 6$ when $x = 0$.

Divide both sides by $(x^2 + 1)$:

$$\frac{dy}{dx} = \frac{4x}{x^2 + 1}$$

Integrate both sides with respect to x :

$$y = 2 \int \frac{2x}{x^2 + 1}$$

Writing the quotient
in the form $\frac{f'(x)}{f(x)}$.

We can use brackets
because $x^2 + 1 > 0$.

$$y = 2 \ln(x^2 + 1) + c$$

Substitute $x = 0$ and $y = 6$:

$$6 = 2\ln(1) + c$$

$$c = 6$$

The particular solution is therefore:

$$y = 2\ln(x^2 + 1) + 6$$



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Differential equations that can be arranged in the form

$$f(y) \frac{dy}{dx} = g(x)$$

can be solved by the method of **separating the variables**.

This method works by collecting all the terms in y , including the ' dy ', on one side of the equation, and all the terms in x , including the ' dx ', on the other side, and then integrating.

$$\int f(y) dy = \int g(x) dx$$

Although the dy and the dx have been separated it is important to remember that $\frac{dy}{dx}$ is not a fraction.

For example, avoid writing:

$$f(y) dy = g(x) dx$$



Separable variables

Here is an example:

Find the general solution to $\frac{dy}{dx} = \frac{x+2}{y}$.

Rearrange to give: $y \frac{dy}{dx} = x + 2$

Separate the variables and integrate:

$$\int y \, dy = \int (x + 2) \, dx$$

We only need a 'c' on one side of the equation.

$$\rightarrow \frac{y^2}{2} = \frac{x^2}{2} + 2x + c$$

$$y^2 = x^2 + 4x + A$$

$$y = \sqrt{x^2 + 4x + A}$$

You can miss out the step

$$\int y \frac{dy}{dx} \, dx = \int (x + 2) \, dx$$

and use the fact that

$$\dots \frac{dy}{dx} \, dx = \dots dy$$

to separate the dy from the dx directly.



Find the particular solution to the differential equation

$$\frac{dy}{dx} = e^{3x-y}$$

given that $y = \ln \frac{7}{3}$ when $x = 0$.

Using the laws of indices this can be written as:

$$\frac{dy}{dx} = e^{3x} e^{-y}$$

Separating the variables and integrating with respect to x gives:

$$\int e^y dy = \int e^{3x} dx$$

$$e^y = \frac{1}{3} e^{3x} + c$$

Take the natural logarithms of both sides:

$$y = \ln\left(\frac{1}{3} e^{3x} + c\right)$$



Given that $y = \ln \frac{7}{3}$ when $x = 0$:

$$\ln \frac{7}{3} = \ln\left(\frac{1}{3} + c\right)$$

$$c = 2$$

The particular solution is therefore:

$$y = \ln\left(\frac{1}{3} e^{3x} + 2\right)$$



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Remember, the rate of change of one variable, say s , with respect to another variable, t , is $\frac{ds}{dt}$.

Many real-life situations involve the rate of change of one variable with respect to another.

Since these situations involve derivatives they are modelled using differential equations.

For example, suppose we hypothesize that the rate at which a particular type of plant grows is proportional to the difference between its current height, h , and its final height, H .

The word “rate” in this context refers to the change in height with respect to time. We can therefore write:

$$\frac{dh}{dt} \propto (H - h)$$



We can write this relationship as an equation by introducing a positive constant k :

$$\frac{dh}{dt} = k(H - h)$$

The general solution to this differential equation can be found by separating the variables and integrating.

Remember the minus sign, because we have $-h$. (H is a constant).

$$\int \frac{1}{H - h} dh = \int k dt$$

$$-\ln(H - h) = kt + c$$

$$\ln(H - h) = -kt + c$$

$$H - h = e^{-kt+c}$$

$$h = H - e^{-kt} e^c$$

$$h = H - Ae^{-kt} \text{ where } A = e^c$$

This is the general solution to the differential equation:

$$\frac{dh}{dt} = k(H - h)$$

If we are given further information then we can determine the value of the constants in the general solution to give a particular solution.

For example, suppose we are told that the height of a plant is 5 cm after 7 days and that its final height is 20 cm.

We can immediately use this value for H to write:

$$h = 20 - Ae^{-kt}$$

Also, assuming that when $t = 0$, $h = 0$:

$$0 = 20 - A$$

$$A = 20$$



And finally using the fact that when $t = 7$, $h = 5$:

$$5 = 20 - 20e^{-7k}$$

$$20e^{-7k} = 15$$

$$e^{-7k} = \frac{3}{4}$$

Take the natural logarithms of both sides:

$$-7k = \ln\left(\frac{3}{4}\right)$$

$$k = -\frac{\ln\left(\frac{3}{4}\right)}{7}$$

This gives the particular solution:

$$h = 20 - 20e^{\frac{t \ln \frac{3}{4}}{7}}$$

Find the height of the plant after 21 days.

Using $t = 21$ in the particular solution gives

$$\begin{aligned}h &= 20 - 20e^{3\ln(\frac{3}{4})} \\&= 20 - 20(\frac{3}{4})^3 \\&= 11\frac{9}{16} \text{ cm}\end{aligned}$$

Using the fact that
 $e^{3\ln(\frac{3}{4})} = (\frac{3}{4})^3$

Comment on the suitability of this model as the plant reaches its final height.

Using this model the plant will reach its final height when:

$$e^{\frac{t\ln\frac{3}{4}}{7}} = 0 \quad \leftarrow \text{This will never happen.}$$

Since e^x never equals 0 this model predicts that the plant will get closer and closer to its final height without ever reaching it.



The most common situations that are modelled by differential equations are those involving exponential growth and decay.

Remember, exponential growth occurs when a quantity *increases* at a rate that is proportional to its size.

For example, suppose that the rate at which an investment grows is proportional to the size of the investment, P , after t years.

We can write this as: $\frac{dP}{dt} \propto P$

This gives us the differential equation:

$$\frac{dP}{dt} = kP$$

where k is a positive constant.

If the initial investment is £1000 and after 5 years the balance is £1246.18, find the particular solution to this differential equation.

$$\frac{dP}{dt} = kP$$

$$\frac{1}{P} \frac{dP}{dt} = k$$

Integrating both sides with respect to t gives:

$$\int \frac{1}{P} dP = \int k dt$$

$$\ln P = kt + c$$

We don't need to write $|P|$ because $P > 0$.

$$P = e^{kt+c}$$

$$P = e^{kt} e^c$$

Exponential growth

$$P = Ae^{kt} \text{ where } A = e^c$$

This is the general solution to $\frac{dP}{dt} = kP$.

Now, using the fact that when $t = 0$, $P = 1000$:

$$1000 = Ae^0$$

$$A = 1000$$

Also when $t = 5$, $P = 1246.18$:

$$1246.18 = 1000e^{5k}$$

$$e^{5k} = 1.24618$$

$$5k = \ln 1.24618$$

$$k = 0.044 \text{ (to 3 s.f.)}$$



The particular solution is therefore:

$$P = 1000e^{0.044t}$$

Find the value of the investment after 10 years.

When $t = 10$:

$$P = 1000e^{0.44}$$

$$P = \text{£}1552.71$$

How long will it take for the initial investment to double?

Substitute $P = 2000$ into the particular solution:

$$2000 = 1000e^{0.044t}$$

$$0.044t = \ln 2$$

$$t = \frac{\ln 2}{0.044}$$

$$\approx 15.75 \text{ years}$$



Remember, exponential decay occurs when a quantity *decreases* at a rate that is proportional to its size.

For example, suppose the rate at which the concentration of a certain drug in the bloodstream decreases is proportional to the amount of the drug, m , in the bloodstream at time t .

Since the rate is decreasing we write:

$$-\frac{dm}{dt} \propto m$$

This gives us the differential equation:

$$\frac{dm}{dt} = -km$$

where k is a positive constant.



Separating the variables and integrating gives:

$$\int \frac{1}{m} dm = \int -k dt$$

$$\ln m = -kt + c$$

$$m = e^{-kt+c}$$

$$m = e^{-kt} e^c$$

$$m = Ae^{-kt} \quad \text{where } A = e^c$$

This is the general solution to the differential equation $\frac{dm}{dt} = -km$.

Suppose a patient is injected with 5 ml of the drug.



There is 4 ml of the drug remaining in the patient's bloodstream after 1 hour. How long after the initial dose is administered will there be only 1 ml remaining?

The initial dose (when $t = 0$) is 5 ml and so we can write directly:

$$m = 5e^{-kt}$$

Also, given that $m = 4$ when $t = 1$ we have:

$$4 = 5e^{-k}$$

$$e^{-k} = \frac{4}{5}$$

$$-k = \ln\left(\frac{4}{5}\right)$$

This gives us the particular solution:

$$m = 5e^{t \ln\left(\frac{4}{5}\right)}$$

We could also write this as

$$m = 5\left(\frac{4}{5}\right)^t$$

When $m = 1$ we have:

$$1 = 5e^{t \ln(\frac{4}{5})}$$

$$e^{t \ln(\frac{4}{5})} = \frac{1}{5}$$

$$t \ln(\frac{4}{5}) = \ln(\frac{1}{5})$$

$$t = \frac{\ln(\frac{1}{5})}{\ln(\frac{4}{5})}$$

$$t \approx 7.2$$

So it will be about 7 hours and 12 minutes before the amount of drug in the bloodstream reduces to 1 ml.

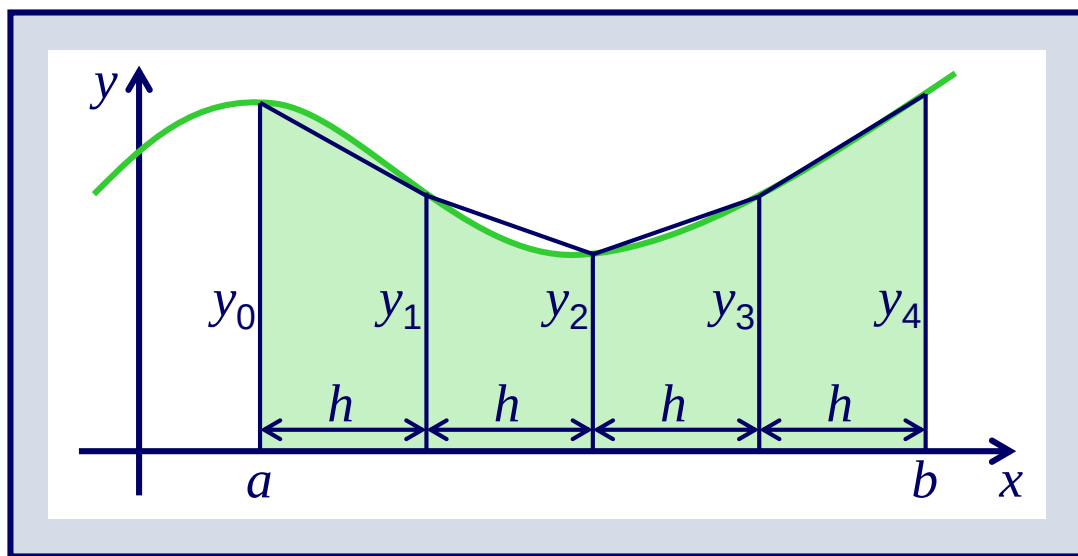


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The trapezium rule

Previously in the course we used the trapezium rule as a method for approximating the area under a curve. Suppose, for example, that we wish to find the area under a curve, $y = f(x)$, between $x = a$ and $x = b$.



We can divide the area into four trapeziums of equal width, h .

The parallel sides of the four trapeziums are given by the five ordinates y_0, y_1, y_2, y_3 and y_4 .

In general, if there are n trapeziums there will be $n + 1$ ordinates.

The trapezium rule

The approximate area using the trapezium rule is:

$$\begin{aligned}\int_a^b f(x)dx &\approx \frac{1}{2}h(y_0 + y_1) + \frac{1}{2}h(y_1 + y_2) + \frac{1}{2}h(y_2 + y_3) + \frac{1}{2}h(y_3 + y_4) \\ &= \frac{1}{2}h(y_0 + y_1 + y_1 + y_2 + y_2 + y_3 + y_3 + y_4) \\ &= \frac{1}{2}h(y_0 + 2y_1 + 2y_2 + 2y_3 + y_4)\end{aligned}$$

The ordinates have to be spaced out evenly so that the width of each trapezium is the same.

$\therefore$ For n trapeziums of equal width h : $h = \frac{b-a}{n}$

In general, the trapezium rule with n trapeziums is:

$$\int_a^b f(x)dx \approx \frac{1}{2}h(y_0 + 2y_1 + 2y_2 + \dots + 2y_{n-1} + y_n)$$

The trapezium rule

Use the trapezium rule with four trapeziums to estimate the value of

$$I = \int_0^2 e^{-2x} dx$$

to 3 significant figures.

By calculating the actual value of I , find the percentage error given using the trapezium rule with four trapeziums.

The width h of each trapezium = $\frac{2-0}{4} = \frac{1}{2}$

Using a table to record the value of each ordinate to 3 s.f.:

x	0	$\frac{1}{2}$	1	$\frac{3}{2}$	2
$y = e^{-2x}$	1	e^{-1}	e^{-2}	e^{-3}	e^{-4}

 y_0  y_1  y_2  y_3  y_4

The trapezium rule

We can now work out the area using

$$\int_a^b f(x)dx \approx \frac{1}{2}h(y_0 + 2y_1 + 2y_2 + \dots + 2y_{n-1} + y_n)$$

with $h = \frac{1}{2}$ and the ordinates given by the table.

$$\begin{aligned}\int_0^2 e^{-2x} dx &\approx \frac{1}{4}(1 + 2(e^{-1}) + 2(e^{-2}) + 2(e^{-3}) + e^{-4}) \\ &= 0.531 \text{ (to 3 s.f.)}\end{aligned}$$

We can find the actual value of I using integration.

$$\begin{aligned}\int_0^2 e^{-2x} dx &= \left[-\frac{1}{2}e^{-2x}\right]_0^2 \\ &= -\frac{1}{2}e^{-4} + \frac{1}{2}e^0 \\ &= 0.491 \text{ (to 3 significant figures)}\end{aligned}$$



Calculating the percentage error

The percentage error is given by:

$$\frac{\text{estimated value} - \text{actual value}}{\text{actual value}} \times 100\%$$

This gives us the percentage error:

$$\frac{0.531 - 0.491}{0.491} \times 100\% = 8.15\%$$

This percentage is fairly large.

Remember that a greater degree of accuracy can be achieved by using more trapeziums.



- Using trigonometric identities in integration
- Using partial fractions in integration
- First-order differential equations
- Differential equations with separable variables
- Using differential equations to model real-life situations
- The trapezium rule
- **Examination-style questions**



Examination-style question 1

$$f(x) = \frac{7x+11}{(4-x)(3x+1)}$$

a) Find the value of the constants A and B such that:

$$f(x) = \frac{A}{4-x} + \frac{B}{3x+1}$$

b) The region R is bound by the curve $y = f(x)$, the coordinate axes and the line $x = 2$.

Find the area of the region R , writing your solution in the form $\ln a$, where a is given to 3 significant figures.

a) Let
$$\frac{7x+11}{(4-x)(3x+1)} \equiv \frac{A}{4-x} + \frac{B}{3x+1}$$



Examination-style question 1

Multiplying through by $(4 - x)(3x + 1)$:

$$7x + 11 \equiv A(3x + 1) + B(4 - x) \quad \textcircled{1}$$

Substituting $x = 4$ into $\textcircled{1}$:

$$39 = 13A$$

$$A = 3$$

Substituting $x = -\frac{1}{3}$ into $\textcircled{1}$:

$$-\frac{7}{3} + 11 = B(4 + \frac{1}{3})$$

$$\frac{26}{3} = \frac{13}{3} B$$

$$B = 2$$



Examination-style question 1

b) The area of the region R is given by:

$$\begin{aligned}\int_0^2 \frac{7x+11}{(4-x)(3x+1)} dx &= \int_0^2 \left(\frac{3}{4-x} + \frac{2}{3x+1} \right) dx \\&= \left[-3\ln(4-x) + \frac{2}{3}\ln(3x+1) \right]_0^2 \\&= -3\ln(2) + \frac{2}{3}\ln(7) + 3\ln(4) - \frac{2}{3}\ln 1 \\&= \ln(2^{-3}) + \ln(7^{\frac{2}{3}}) + \ln(4^3) \\&= \ln\left(\frac{7^{\frac{2}{3}} \times 4^3}{2^3} \right) \\&= \ln(29.3)\end{aligned}$$



Examination-style question 2

Water is leaking from the bottom of a tank of height 1 m. The rate at which the water leaks from the tank is given by the differential equation.

$$\frac{dh}{dt} = -A\sqrt{h}$$

where h is the depth of the water in the tank at time t and A is a positive constant.

- Solve this differential equation given that the tank is initially filled to full capacity.
- After three hours the depth of the water is 0.25 m. Find the depth of the water in the tank after four hours.
- How long does it take for the tank to empty from full capacity?



Examination-style question 2

a)
$$\frac{dh}{dt} = -A\sqrt{h}$$

Separating the variables and integrating gives:

$$\int \frac{1}{\sqrt{h}} dh = \int -A dt$$

$$2\sqrt{h} = -At + c$$

$$\sqrt{h} = \frac{c - At}{2}$$

$$h = \left(\frac{c - At}{2} \right)^2$$



Examination-style question 2

Given that when $t = 0$, $h = 1$:

$$1 = \left(\frac{c}{2} \right)^2$$

$$\frac{c}{2} = 1$$

$$c = 2$$

The height of the water in the tank is therefore given by:

$$h = \left(\frac{2 - At}{2} \right)^2$$

$$= \left(1 - \frac{At}{2} \right)^2$$

Examination-style question 2

b) When $t = 3$, $h = 0.25$ so:

$$0.25 = \left(1 - \frac{3A}{2}\right)^2$$

$$0.5 = 1 - \frac{3A}{2}$$

$$\frac{3A}{2} = 0.5$$

$$A = \frac{1}{3}$$

$\therefore$ The particular solution is:

$$h = \left(1 - \frac{\frac{1}{3}t}{2}\right)^2$$
$$= \left(1 - \frac{t}{6}\right)^2$$

Examination-style question 2

After four hours, when $t = 4$:

$$h = \left(1 - \frac{4}{6}\right)^2$$

$$= \left(\frac{1}{3}\right)^2$$

$$= \frac{1}{9}$$

$\therefore$ The height of the water after 4 hours is **0.11 m** (to 2 d.p.).



Examination-style question 2

c) The tank will be empty when $h = 0$, that is when:

$$0 = \left(1 - \frac{t}{6}\right)^2$$

$$1 - \frac{t}{6} = 0$$

$$\frac{t}{6} = 1$$

$$t = 6$$

∴ The tank will take **6 hours** to empty from full capacity.

